Introduction to Programming

Lecture 17

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String Handling

String Manipulation Functions

Character

ASCII

1 byte = 8 bits

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Example 1

```
#include<iostream.h>
main ( )
{
      int i ;
      char c ;
      for( i = 0; i < 256 ; i ++ )
      {
       c = i ;
       cout << i << "\t" << c <<endl ;
      }
}
```

ctype Functions

int isdigit (int c)
int isalpha (int c)
int isalnum (int c)
int isxdigit (int c)
int islower (int c)
int isupper (int c)
int tolower (int c)
int toupper (int c)

int isspace (int c)
int iscntrl (int c)
int ispunct (int c)
int isprint (int c)
int isgraph (int c)

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isdigit ( ) Function

```
int isdigit ( int c ) ;
```

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isalpha ( ) Function

```
int isalpha ( int c ) ;
```

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isalnum ( ) Function

```
int isalnum ( int c ) ;
```

islower ( ) Function

```
int islower ( int c ) ;
```

isupper ( ) Function

```
int isupper ( int c ) ;
```

tolower ( ) Function

```
int tolower ( int c ) ;
```

toupper ( ) Function

```
int toupper ( int c ) ;
```

getchar ( ) ;

```
cout << "Please enter a character string then press enter";
while ( ( c = getchar ( ) ) != '\n' )
{
   if ( islower ( c ) )
     lc ++ ;
   else if ( isupper ( c ) )
     uc ++ ;
   else if (isdigit ( c ) )
     dig ++;
   else if ( isspace ( c ) )
     ws ++ ;
   else if ( ispunct ( c ) )
     pun ++ ;
   else
     oth ++ ;
}
```

String Conversion Functions

double atof (char * str)

int atoi (char * str)

long atol (char * str)

atoi ( ) Function

```
char str [ ] ;
int age ;
age = atoi ( str ) ;
```

atof ( ) Function

12.89

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atof ( ) Function

```
char str [ ] ;
double dVar ;
dVar = atof ( str ) ;
```

int main (int agrc, char **agrv)

‘argc’ stands for a count of the number of arguments

‘argv’ stands for a vector of arguments

String Functions

String Manipulation Functions

Function prototype	Function description
char *strcpy(char *s1, const char *s2)	Copies string s2 into array s1. The value of s1 is returned.
char *strncpy(char *s1, const char *s2, size_t n)	Copies at most n characters of string s2 into array s1. The value of s1 is returned.
int strcmp(const char *s1, const char *s2);	Compares string s1 to s2 Returns a negative number if s1 < s2, zero if s1 == s2 or a positive number if s1 > s2
int strncmp(const char *s1, const char *s2, size_t n);	Compares up to n characters of string s1 to s2 Returns values as above
char *strcat(char *s1, const char *s2)	Appends string s2 to array s1. The first character of s2 overwrites the terminating null character of s1. The value of s1 is returned.
char *strncat(char *s1, const char *s2, size_t n)	Appends at most n characters of string s2 to array s1. The first character of s2 overwrites the terminating null character of s1. The value of s1 is returned.

```
int sum ;
int sum_even ;
int sum_odd ;

myStrcpy ( ) ;
```

String Manipulation Functions

```
char * strcpy (char *s1 , const char *s2 ) ;

char * strncpy ( char *s1 , char *s2 , int n ) ;

char * strcat (char *s1 , char *s2 ) ;

char * strncat ( char *s1 , char *s2 , int n ) ;
```

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strcmp ( ) Function

```
int strcmp (const char *s1 , const char *s2 ) ;
```

strncmp ( ) Function

```
int strncmp ( const char *s1 , const char *s2 , int n ) ;
```

strlen ( ) Function

```
int strlen ( const char *s ) ;
```

Search Functions

Search Functions

Function prototype	Function description
char *strchr(const char *s, int c);	Locates the first occurrence of character c in string s. If c is found, a pointer to c in s is returned. Otherwise, a NULL pointer is returned.
size_t strcspn(const char *s1, const char *s2);	Determines and returns the length of the initial segment of string s1 consisting of characters not contained in string s2.
size_t strspn(const char *s1, const char *s2);	Determines and returns the length of the initial segment of string s1 consisting only of characters contained in string s2.
char *strpbrk(const char *s1, const char *s2);	Locates the first occurrence in string s1 of any character in string s2. If a character from string s2 is found, a pointer to the character in string s1 is returned. Otherwise, a NULL pointer is returned.
char *strrchr(const char *s, int c);	Locates the last occurrence of c in string s. If c is found, a pointer to c in string s is returned. Otherwise, a NULL pointer is returned.
char *strstr(const char *s1, const char *s2);	Locates the first occurrence in string s1 of string s2. If the string is found, a pointer to the string in s1 is returned. Otherwise, a NULL pointer is returned.
char *strtok(char *s1, const char *s2);	A sequence of calls to strtok breaks string s1 into "tokens"—logical pieces such as words in a line of text—separated by characters contained in string s2. The first call contains s1 as the first argument, and subsequent calls to continue tokenizing the same string contain NULL as the first argument. A pointer to the current token is returned by each call. If there are no more tokens when the function is called, NULL is returned.

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This is a test

" " wrong

' ' right

NULL